

Yash Maheshwari

yashmahe2018@gmail.com • (650) 208-4653 • Mountain View, CA • yash-maheshwari.com
AI Researcher • Patent Holder • Education + AI Policy Advocate

EDUCATION

Mountain View High School

Aug 2023 – Jun 2027

GPA: 4.0 UW / 4.85 W • Foothill College Dual Enrollment (CS Associate Degree track, 3.96 GPA)

RESEARCH EXPERIENCE

Research Intern, Lemons Lab | [Stanford University \(Prof. Chris Lemons\)](#)

Jun 2025 – Present

- Built the AIOps evaluation suite for Kai, an AI reading tutor used by 1,200+ students and 50+ teachers in 10+ districts
- Introduced Kai to dozens of educators, administrators, and parents at the Annual Stanford Down Syndrome Conference
- Helped take PAWS, a kindergarten handwriting project, from web prototype to native iPad app (handwriting recognition, AI grading, voice-guided instruction); now being prepared for classroom testing
- Cut latency 75% via prompt and model A/B tests, then built a non-llm pathway answering in under 50 ms at full accuracy

Research Intern, Shah Lab | [Stanford University \(Prof. Nigam Shah\)](#)

Aug 2026 – Present

- Maintain HealthAdminBench, a benchmark for LLM agents on healthcare admin work (prior authorizations, denial appeals, DME orders); improved its architecture and A/B tested models with engineers from AI labs including Anthropic
- Building HealthAdminBench V2 on the Harbor framework: an Epic EHR environment, a larger task set, a new set of frontier models, and expanded error tracing of where and how each model gets stuck

Researcher, Subquadratic Language Models | [Independent](#)

Apr 2026 – Present

- Designed BiRWKV-7, a ~28M-parameter recurrent language model; accepted to the BabyLM Workshop at EMNLP 2026, Budapest
- Reached 69.5% BLiMP grammatical accuracy with linear-time $O(T)$ training at 23% of the parameters of rival transformers, trained entirely on Apple M5 hardware with MLX

Researcher, Sub-1-Bit Quantization | [Independent](#)

Jul 2026 – Present

- Submitted two papers to NeurIPS 2026 workshops (AXIOM, ODI) on language models below one bit per weight
- Showed that retraining only 8% of a compressed model recovers 97% of lost quality, and that natively trained models are more robust to noisy input

SLM Researcher, Math Tutoring | [Independent / JEI Publication](#)

Jan 2025 – Nov 2025

- Accepted to the Journal of Emerging Investigators: fine-tuned Llama 3 math tutors across parameter sizes and epochs
- Found that parameter count, not epoch count, drives accuracy; the 8B model reached 93.7% step accuracy
- Presented the work at the 2025 Santa Clara County Science Fair

PATENTS

- HIERARCHICAL AGGREGATION TREE FOR MCP SERVER SELECTION AND EXECUTION (nonprovisional, filed Oct 2025)
- PREDICTIVE COMPLIANCE FOR AI AGENTS (provisional, filed Oct 2025)

WORK EXPERIENCE

AI Research Contractor | [Kinetic Systems and Technologies](#)

Jul 2026 – Aug 2026

- Wrote healthcare workflow tasks for frontier AI lab clients that current models could not yet solve
- Traced model error paths on each task to document where and how the models failed
- Quality-controlled existing tasks for accuracy and realism before delivery to clients

AI Engineering Intern | [Aisera Inc.](#)

Jun 2025 – Aug 2025

- Built MCP servers (deal scoring, tenant health, Clari call insights, ideal customer profile) on Salesforce, Clari, and Slack data; used by the sales team in product demos
- Developed an MCP Bridge supporting both legacy HTTP and SSE connections
- Won 2nd place in the company-wide AI hackathon against 30 teams of engineers

Executive Board Member | [MVHS Principal's Tech Internship](#)

Sep 2024 – Present

- Joined at the program's launch, on the Executive Board since Jun 2025; work with Dr. Kip Glazer to grow the program

- Built and shipped the AI Policy Pathway (helps schools draft an AI policy fit to their community) and Bridge the Gap 360 (simplifies administrator evaluations for teachers)
- Organized and presented at AI & Education Parent Night (60+ attendees), CAL-MSCS Day (80+ STEM teachers statewide), and an AI Playlab (100+ middle schoolers and parents)
- Led three co-design workshops with students, parents, and teachers that produced a draft district AI Philosophy

EngageAI Institute Intern | [Vanderbilt University / EngageAI](#)

Feb 2025 – Apr 2025

- Designed two AI Bill of Rights card games (Student and Educator editions) and an adventure game with Dr. Sarah Burriss
- Wrote a six-point Student Bill of Rights for classroom AI use; presented the games at the ASU+GSV AI Show and FETC

Co-Founder & CEO | [Tech Spark \(501c3\) / Novus Innovation LLC](#)

2020 – Present

- Ran summer programs for 5 consecutive years teaching robotics and software development to dozens of K-8 students
- Designed, crowdfunded, and made the card game Battle Timz; sold 300+ copies on Amazon, Kickstarter, and door-to-door

AI POLICY & ADVOCACY

- Met with 10 legislative offices across 9 states on AI bills, including FL SB 482 (AI Bill of Rights), ME LD 2162 (human-like AI), MI SB 760 (companion chatbots), PA SB 939 (AI regulatory sandbox), and CO HB26-1139 (AI in health care)
- Briefed U.S. Rep. Sam Liccardo's office on the student view of AI in schools; emailed 1,500+ state legislators on 75+ bills
- Co-designed the local school district's AI Philosophy through community workshops; covered by The Washington Post as a national model for student-led district AI policy, and by the Los Altos Town Crier
- **MVLA EdTech Committee:** Set district AI goals for instruction and responsible adoption (Oct 2025 – Present)
- **Principal's Advisory Council:** Member since Aug 2025, lead since Jun 2026; pushed for an AP course cap and other student stress measures; now on student campus resources, grade inequity, and AI in the classroom.

SPEAKING & MEDIA

- **Common Sense Media Summit 2026 (San Francisco):** Two keynote panels, opening and closing day one, before 600+ leaders in tech, education, philanthropy, and policy
- **FETC 2026 (Orlando):** Led two sessions, "Teaching AI Ethics Through Play" and "Student-Led Innovation: Rethinking Tech Internships in K-12 Education"
- **ASU+GSV Summit (San Diego):** Presented the student co-design model in "Learners Light the Way"; one of 3 high schoolers at the Walton Family student breakfast; raised nearly \$10K for the trip
- **Keynote FETC 2027 (Orlando):** Will keynote the conference in front of 9,000+ educators, students, and administrators.
- **KCI and Google HQ:** Presented the Tech Internship model at Foothill's Krause Center for Innovation; AI-in-education panelist with Google engineers and designers
- **Press:** Interviewed by the Center for Digital Education, Thinkering Collective, and Amplify on student views of AI

AWARDS & RECOGNITION

- 1st Place, MVHacks (custom-trained Llama 3.2 3B model)
- 2nd Place, Aisera Internal AI Hackathon (30 engineering teams)
- Honorable Mention, 2024 Congressional App Challenge (Rep. Anna Eshoo)
- Highest Rookie Team Award, FRC World Championship, Houston 2024 (Software Lead, Team 9584)
- Judges' Choice Award, FTC South Bay League, 2024-25 season (team founder and mentor)

LEADERSHIP & EXTRACURRICULARS

- **RL Game Club:** Founder and President (Sep 2025 – Jun 2026); taught RL through CartPole, Tic-Tac-Toe, and Connect 4 and ran model competitions for a dozen+ members
- **FTC Robotics:** Founded and mentor a middle school team (Feb 2024 – Present); placed 5th of ~20 teams in Oct 2025
- **FRC Robotics:** Software Lead, Team 9584 (Jun 2023 – Jan 2025); 17th at SF Regional, 27th of 75 in division at Worlds
- **Summer Programs:** LaunchX (2024); Brown Codes, Ciphers & Cryptography (2023); Dartmouth GenCyber (2023)
- Cross-country • Pianist • 3 published iOS apps on the App Store

TECHNICAL SKILLS

Languages: Python, Swift, TypeScript, JavaScript **Frameworks & Tools:** PyTorch, MLX, React, Next.js, SwiftUI/Xcode, Git
AI/ML: LLM fine-tuning (Llama/Qwen), subquadratic architectures, sub-1-bit quantization, model evaluation and agent benchmarking, agentic task design, MCP development, RAG, reinforcement learning.